USB-C

J1
USB-C (USB4105)
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
SBU1
SBU2
A9
B9
A5
B5
A7
B7
A6
B6
A8
B8
USB_DM
USB_DP
R14 5.1k
R13 5.1k
SH
GND
D10
SMF5.0A
VBUS_F
F1
1A fast (466)
D4
SS14
+5V
CH1_OUT
GND
CH1_IN
CH2_OUT
VCC
CH2_IN
USB_DM_ESP
USB_DP_ESP
D5
TPD2S017

ESP32-C6 MCU

+3V3
EN
I2S_WS
OC3_OUT
OC2_OUT
MTMS/GPIO4/ADC1_CH4
MTDI/GPIO5/ADC1_CH5
I2S_MCLK
I2S_DOUT
MTCK/GPIO6/ADC1_CH6 U1
MTDO/GPIO7
NC
EN/CHIP_PU
3V3
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
ESP32-C6-WROOM-1-N8
ESP32-C6-WROOM-1
U0TXD/GPIO16
U0RXD/GPIO17
GPIO13/USB_D+
GPIO12/USB_D−
USB_DP_ESP
USB_DM_ESP
GPIO8
GPIO9
BOOT
GPIO10
I2S_DIN
GPIO11
I2S_BCLK
GPIO15
GPIO18
I2C_SDA
GPIO19
I2C_SCL
GPIO20
GATE2_DRV
GPIO21
GATE1_DRV
GPIO22
GATE3_DRV
GPIO23
OC1_OUT
C3 10uF
C6 100nF
R12 3.3k

POWER

U2
SGM2212-3.3
VIN
VOUT
GND
C2 10uF
C4 10uF

POWER LED

D6
PWR
R15 1k

RELAY DRIVERS

GATE1_PRE
R6 100
D1 1N4148W
K1 G6K-2F-Y 4.5V
Q1 2N7002
R9 10k
IN_P4
P2
P3
VIRTUAL PTT
bridges IN_P4<−>P2 (talk);
gate runs via K3 interlock
GATE2_DRV
R4 100
D2 1N4148W
K2 G6K-2F-Y 4.5V
Q2 2N7002
R7 10k
R16 2.2k
DOOR OPENER
bridges P2<−>P3 via R16,
emulating the WF26 OeT button
GATE3_DRV
R5 100
D3 1N4148W
K3 G6K-2F-Y 4.5V
Q3 2N7002
R8 10k
CHIME SUPPRESS
breaks line 4 (IN_P4−>P4);
pole B interlocks K1 gate

BELL SENSE

OC1_IP
OC2_IP
OC3_IP
SW3 CAS-220TB1
SW4 CAS-220TB1
SW5 CAS-220TB1
D7 1N4148W
D8 1N4148W
D9 1N4148W
OK1 LTV-217 (PC817)
OK2 LTV-217 (PC817)
OK3 LTV-217 (PC817)
R17 5.1k
R1 5.1k
R2 5.1k
R21 10k
R22 10k
R23 10k
R3 1k
SESSION ACTIVE (talk relay, P5−>P2)
HOUSE BELL / Tuerruf (P1−>IN_P4)
APARTMENT BELL / Etagenruf (P1−>P5)
WF26 (6-way screw)
Pin_1
Pin_2
Pin_3
Pin_4
Pin_5
Pin_6

RESET / BOOT

R10 10k
C5 1uF
SW2 EN
R11 10k
SW1 BOOT

AUDIO ES8311 + LINE XFMR

R19 10k
C7 100nF
C8 100nF
C9 100nF
C10 10uF
R18 10k
I2C_SCL
I2C_SDA
U3
ES8311
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
I2S_MCLK
I2S_BCLK
I2S_DIN
I2S_WS
I2S_DOUT
ES_CE
ES_VMID
R20 10k
C12 1uF
T1
SM-LP-5001
C14 1uF
C15 1uF
C17 1uF
C16 1uF
C11 10uF
C13 1uF
R24 1k
R25 1k
R27 10k
R26 10k

Gunnar Beutner
Sheet: /
File: doorbell.kicad_sch
Title: Doorbell controller (Klingel V4)
Size: A3
Date: 2026-06-12
Rev: V4
KiCad E.D.A. 10.0.3
Id: 1/1